

Derbyshire Shared Care Pathology Guidelines

Hyponatraemia in Adults

Purpose of Guideline

The investigation and management of adult patients with newly diagnosed hyponatraemia.

Hyponatraemia can be defined as a plasma sodium below 133 mmol/L.

The clinical significance of hyponatraemia depends on its severity, its speed of onset and its underlying cause.



The laboratory usually telephones sodium results < 121 mmol/L to GP practices when open or to Derbyshire Health United when the GP practice is closed.

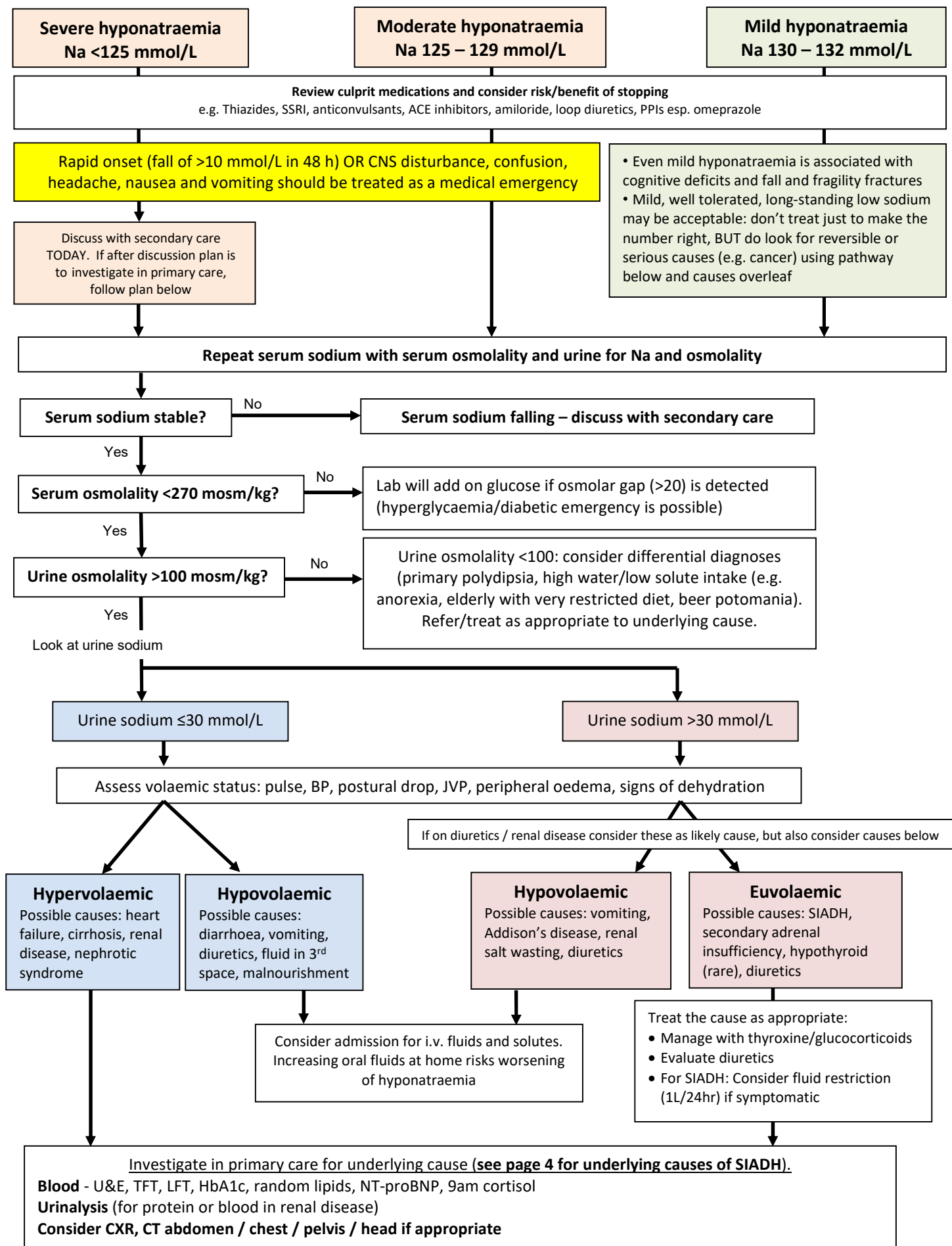
Symptoms

These are primarily neurological and reflect changes in brain volume. They range in severity and include:

- Headache
- Confusion
- Nausea and vomiting
- Lethargy
- Irritability
- Seizures
- Loss of consciousness
- Coma

There is no specific concentration at which they may occur and hyponatraemia developing over days to weeks may be relatively asymptomatic, even with $\text{Na} < 115$ mmol/L. There may be additional symptoms which reflect the underlying cause.

Outline investigation and management of hyponatraemia is shown in the following flow chart with more detailed information in the following text.



When is hyponatraemia a medical emergency?

Hyponatraemia of rapid onset (fall of >10 mmol/L within 48 h) OR if associated with any of the symptoms above may indicate acute hyponatraemic encephalopathy and should be treated as a medical emergency. It is, however, rare in primary care.

Potential causes of acute hyponatraemic encephalopathy include:

- Psychogenic polydipsia
- Other types of excessive habitual water drinking including: excess beer or tea drinkers with limited food intake
- Drugs: ACEi, A2RB, Trimethoprim, diuretics, SSRIs, carbamazepine

What are the main causes of hyponatraemia?

Hyponatraemia is categorised according to extracellular fluid volume status. Clinical assessment of this together with fluid balance information is vital in determining the likely cause. However, clinical assessment may be unreliable and should be aided by biochemical testing.

Excess salt loss:

- Gut: diarrhoea/vomiting, fistula, stoma, etc
- Kidneys: diuretics, ACEi, A2RB, Trimethoprim, adrenal insufficiency

Excess fluid intake:

- Excess inappropriate drinking
- SIADH or excess use of desmopressin
- Hyperglycaemia or hypercalcaemia

Hypervolaemia (oedema states):

- Nephrotic syndrome
- Cardiac failure
- Cirrhotic liver disease

Pseudohyponatraemia:

- Artefactually low sodium due to the presence of grossly elevated levels of triglycerides (eg >30 mmol/L) or protein (eg >100 g/L)
- The laboratory can check for these and measure sodium by an alternative method (direct ion selective electrode) if indicated

Drugs commonly causing hyponatraemia:

- Salt losing diuretics: Thiazide or potassium sparing diuretics
- Carbamazepine
- Sulphonylureas (but not gliclazide)
- Proton pump inhibitors
- Antidepressants, particularly SSRIs
- ACE inhibitors and ARBs
- Opiates
- Trimethoprim

What other information is needed in the hyponatraemic patient?

- Previous sodium results
- Symptoms and signs of neurological disturbance
- Fluid balance information
- Volume status: hypovolaemia, euvolaemia or oedema
- Medical and drug and smoking history

What other investigations are needed in the hyponatraemic patient?

- U&E, serum osmolality, glucose, LFT including protein
- Urine osmolality, urine Na
- Calcium, TFT, Cortisol (9 am), NT-proBNP

Other investigations will depend on cause (e.g. cardiac investigations if CCF, chest X-ray, brain imaging and possibly whole body imaging if SIADH)

How is syndrome of inappropriate ADH (SIADH) diagnosed?

SIADH is a relatively uncommon cause of hyponatraemia. There is no specific test for it and the diagnosis is one of exclusion after other causes have been disproved. The criteria are:

- Normovolaemic
- Normal renal, adrenal, thyroid function
- Urine not maximally dilute (ie, osmolality >100 mosm/kg)
- Urine Na >30 mmol/L
- No drugs causing hyponatraemia, e.g. diuretics/ACEi (5 days off relevant drug)
- Improvement in Na following water restriction

What are the common causes of SIADH?

- Malignancy or ectopic secretion (e.g. small cell Ca bronchus)
- Cerebral (e.g. tumours, infection, CVA, trauma)
- Pulmonary (e.g. pulmonary embolus, pneumonia, trauma, tuberculosis)
- Metabolic (e.g. acute porphyria)

Interpretation of urine results

At this point it is important to review all available results and information from other parts of the assessment (e.g. drug history and previous results) to determine the most likely cause of the hyponatraemia.

Urine osmolality

- ≤ 100 mmol/kg: Consistent with primary water overload, e.g. primary polydipsia, low solute intake
- >100 mmol/kg: Consider urine Na

Urine Na

- ≤ 30 mmol/L: Consistent with low effective arterial blood volume
 - With oedema: CCF, nephrotic syndrome, cirrhosis
 - Without oedema: GI fluid loss, third space loss, previous diuretics, malnourishment
- >30 mmol/L: consistent with renal Na loss
 - Renal loss: Diuretics, kidney disease, adrenal insufficiency, cerebral salt wasting
 - SIADH

Treatment

Treatment of hyponatraemia

Note that: The use of Demeclocycline, urea or vaptans should only be considered on the advice of a specialist

In the absence of acute hyponatraemic encephalopathy, hyponatraemia should be corrected slowly. Mortality is related to the underlying cause, rather than the sodium concentration. There is a risk of central pontine myelinolysis (CPM), also known as osmotic demyelination syndrome, if correction is too rapid. Features of CPM include dysarthria, dysphagia, seizures, altered mental status and quadriparesis, which typically begin 1-3 days after plasma sodium has been corrected.

Chronic, mild, asymptomatic hyponatraemia may not require treatment in its own right, as long as underlying causes have been considered and excluded. Symptomatic patients and those with sodium less than 125 mmol/L need treatment. Stopping hyponatraemic medications can help to confirm the cause of hyponatraemia as well as resolve the problem.

Contacts

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Duty Biochemist	01332 789383 (8am to 7pm, Mon – Fri)
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A&E, MAU and ACC	Via RDH switchboard, 01332 340131(24/7)
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OR

ED Nurse in Charge	07799 337674
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Chesterfield

Duty Biochemist	01246 512212 (9am to 5pm, Mon – Fri)
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Burton

Duty Biochemist	01283 593313 (8.30 am -4pm Mon-Fri), diverts to RDH duty biochemist out of these hours
On Call Consultant Biochemist	Via QHB switchboard, 01283 566333
A&E, EMU and CDU	Via QHB switchboard, 01283 566333

References

1. Clinical practice guideline on diagnosis and treatment of hyponatraemia. Eur J Endocrinol. 2014 Feb 25;170 (3): G1-47. doi: 10.1530/EJE-13-1020.
2. NICE CKS – Hyponatraemia (Last revised July 2025)

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Approval	Date	Name(s)	Designation
Secondary Care UHDB	20.01.26	Dr A Agur	Consultant Endocrinologist
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